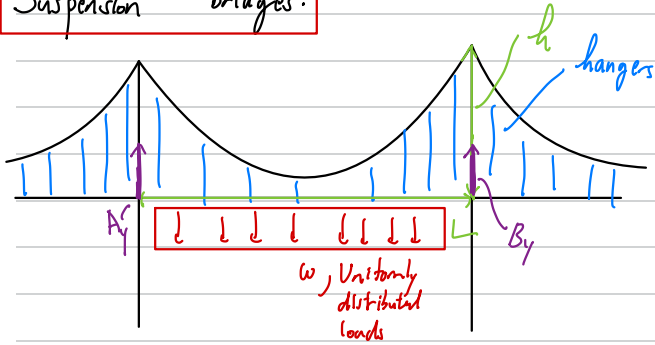


Rooney's CIV102 Finals Notes

True the process $11/12$



Suspension bridges:



Equations:

$$A_y = B_y = \frac{wL}{2}$$

Tension in cable

$$X: T_x (\text{constant}) = \frac{wL^2}{8h}$$

$$Y: T_y (\text{varies}), \text{max} = \frac{wL}{2}$$

$$\text{Max cable force} = \sqrt{\left(\frac{wL^2}{8h}\right)^2 + \left(\frac{wL}{2}\right)^2}$$

Stress / Strain: (F in N, distances in mm)

$$\text{Stress, } \sigma = \frac{F}{A} \text{ (MPa)}$$

$$\text{Strain } \epsilon = \frac{\Delta l}{L_0} \text{ (unitless)}$$

$$\text{Young's modulus, } E = \frac{\sigma}{\epsilon} \text{ (MPa)}$$

$$\text{Axial stiffness, } k = \frac{F}{\Delta l} \text{ (N/mm)}$$

$$F = \frac{AE}{L_0} \Delta l, \quad k = \frac{AE}{L_0}$$

$$\text{or } \Delta l = \frac{FL_0}{EA}$$

Graphs:

$\sigma - \epsilon$ curve (Steel)



$$\text{Energy density, } U = \text{Area under } \sigma/\epsilon \text{ curve}$$

$$= \frac{1}{2} \sigma \epsilon \text{ (linear)}$$

$$\text{or } = \sigma \epsilon \text{ (plateau)}$$

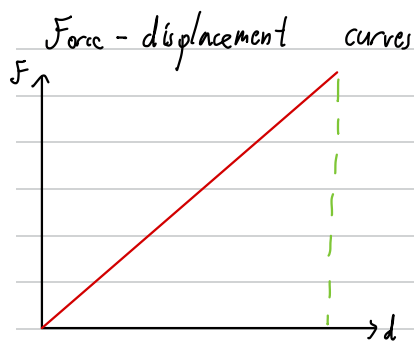
If σ is in MPa, ϵ is in mm/mm
 U is in MJ/m³

$$\text{Energy stored} = U \cdot V_0$$

$$V_0 = A \cdot L_0 \text{ for constant cross sections}$$

- ① linear elastic
- ② Yield plateau
- ③ Strain hardening

↳ ②, ③ = plastic damage → permanent.



Resilience and toughness:

Resilience: Maximum amount of energy a material can absorb before plastic deformation (MJ/m^3)
(Area under linear-elastic region of σ - ϵ curve)

Toughness: Maximum amount of energy a material can absorb before breaking/rupturing (MJ/m^3)
(Area under whole σ - ϵ curve)

Strain energy = area under curve

$$= \frac{1}{2} k (\Delta l)^2$$

Frequency

Natural frequency of vibration, $f_n = \frac{15.76}{\sqrt{\Delta_0}}$, Δ_0 in mm (Given $f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$, $k = \frac{mg}{\Delta_0}$)

Period = $\frac{1}{f_n}$

$f_n = \frac{17.76}{\sqrt{\Delta_0}}$, for trusses

FOS

Failure $\Leftrightarrow \sigma_{\text{demand}} \geq \sigma_{\text{capacity}}$

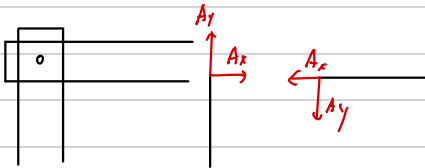
$\sigma_{\text{allowable}} = \frac{\sigma_{\text{fail}}}{\text{FOS}}$ $\leftarrow$ Finding maximum allowable stress given FOS, σ_{fail}

$\text{FOS} = \frac{\text{capacity}}{\text{demand}}$ $\leftarrow$ Finding FOS given capacity (σ_{max}), demand (σ_{demand})

Connection, support types

Connections

a) Hinge
 $M = 0$
 $F_{x,y} \neq 0$



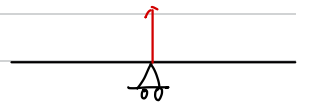
b) Fixed connection
 $M, F_{x,y} \neq 0$

Supports

a) Fixed end ($F_{x,y}, M \neq 0$)



c) Roller ($M, F_x = 0, F_y \neq 0$)



b) Pin ($M = 0, F_{x,y} \neq 0$)



Equations

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_A = 0$$

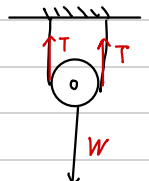
Statically determinate?

a) # of $R_x <$ # of equations
 $\hookrightarrow$ Not static / safe

c) # of $R_x >$ # of equations
 $\hookrightarrow$ indeterminate

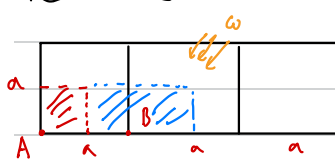
b) # of $R_x =$ # of equations
 $\hookrightarrow$ Determinate

Pulleys?



Trusses

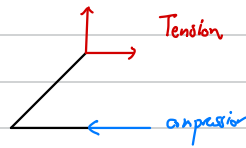
Joint loads



Load at A:
 $= w \cdot \left(\frac{a}{2}\right)^2$

Load at B:
 $= w \cdot \left(\frac{a}{2}\right)(a)$

Member (Tension or compression?)



Is a bridge safe?

1. Select geometry
2. Choose loads (w or wind)
3. Solve for R_x and member force
4. Check if members can resist load (or choose safe member)
5. Find deflection @ midspan
6. Find load by dynamic effects

Moment of inertia, I

$$M = (m y^2) \alpha = I_m \alpha$$

$\hookrightarrow y$ = length of rod, α = angular acceleration

$$I_m \text{ moment of inertia} = \rho \cdot t \cdot \int y^2 dA$$

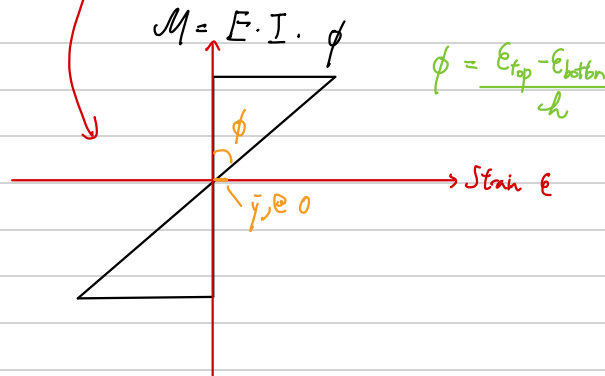
$$I_{\text{rectangle}} = \frac{b h^3}{12} \text{ (mm}^4\text{)}$$

Curvature, ϕ

$$\phi = \frac{d\theta}{dx} \text{ or } \frac{1}{R}$$

$$\epsilon(y) = \phi y, y = \text{distance above } \bar{y}$$

$$M = E \cdot I \cdot \phi$$



Euler's buckling

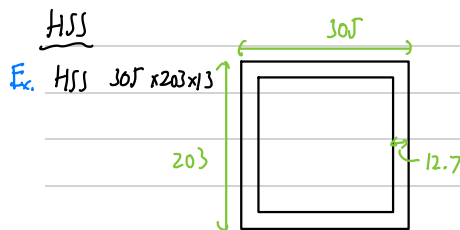
$$P_{\text{crit}} = \frac{\pi^2 E \cdot I}{L^2} \text{ } P \text{ is a compressive force}$$

HSS and designing

$$\sigma_{\text{yield}} = 350 \text{ MPa}$$

$$E = 200,000 \text{ MPa}$$

Designing



$$I_{\text{HSS}} = \frac{\text{bott } h_{\text{out}}^3}{12} - \frac{\text{bot } h_{\text{in}}^3}{12}$$

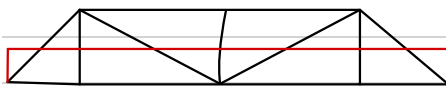
	A	I	r
Tens.		No buckling	$r \geq \frac{L}{200}$
Comp.	$A \geq 2.0 \frac{F}{\sigma_y}$	$I \geq \frac{30 F_{\text{max}} L^2}{\pi^2 E}$	

Wind bracing

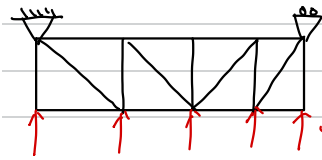
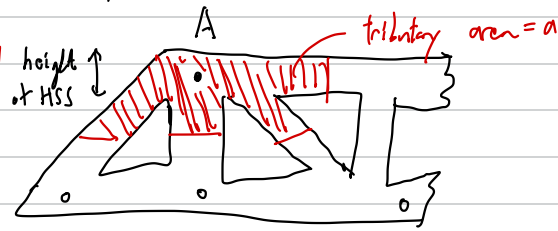
$$w_{wind} \sim 2.0 \text{ kPa} \sim 2.0 \times 10^{-3} \text{ N/mm}^2$$

Tributary area w/o hand rails:

Tributary area w/ hand rails



Side view



Point loads
= (tributary area) w

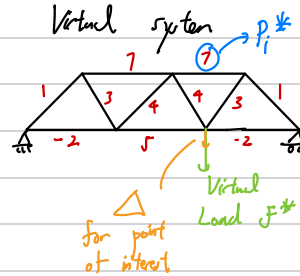
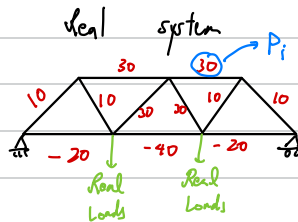
Top/bottom view

Method of Virtual Work

$$F^* \Delta = \sum_{i=1}^n P_i^* \left[\frac{P_i L_i}{E_i A_i} \right]$$

Typically $F^* = 1 \text{ kN}$

$$\Delta = \sum_{i=1}^n P_i^* \left[\frac{P_i L_i}{E_i A_i} \right]$$



(all numbers are made up)

Δ , real displacement

P_i^* , force in member as a result of virtual force (+ve tenj -ve comp)

P_i , initial force in member in real system (+ve tenj -ve comp)

L_i , length of member (same in both)

E_i , Young's Modulus of member (same in both)

A_i , Area of member (same in both)

Dynamic amplification factors

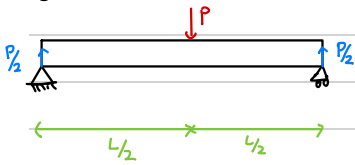
$$DAF = \frac{1}{\sqrt{\left(1 - \left(\frac{f}{f_n}\right)^2\right)^2 + \left(\frac{2\beta f}{f_n}\right)^2}}$$

f is the forced frequency (given usually)
 f_n is the natural frequency = $\frac{17.26}{\sqrt{\Delta_0}}$
 β damping ratio $\sim (0 \pm 0.05)$ (given usually)

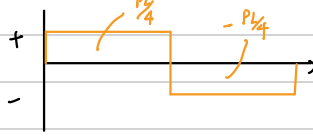
$$W_{dynamic} = \underbrace{W_{stationary}}_{\text{Load under static case}} + DAF \underbrace{(w_0)}_{\text{live (typically given)}}$$

Common FBD, SFD, BMD

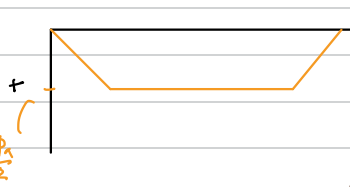
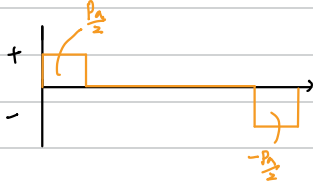
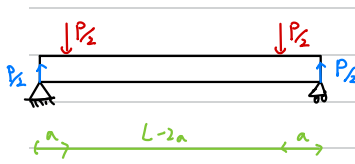
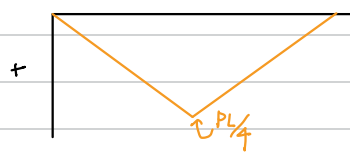
FBD



SFD



BMD

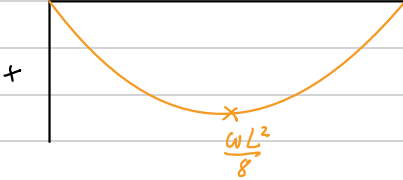
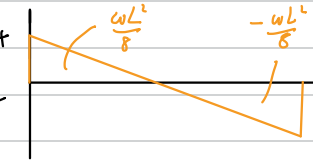
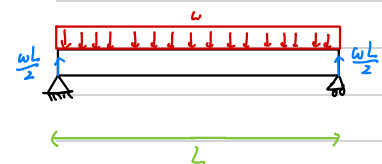
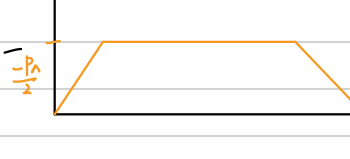
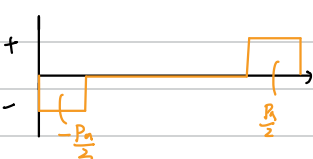
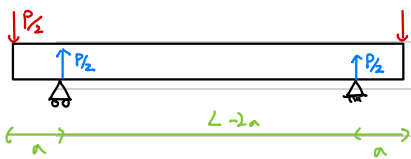


Note:

But in tension
+ve M → Top in comp.



But in comp
-ve M → Top in tension



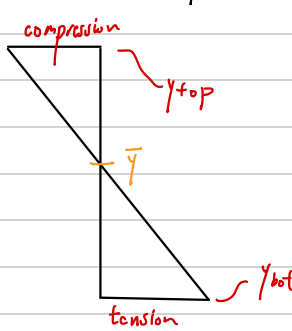
Bending stress

Navier's equation

$$\sigma = \frac{My}{I}$$

from BMD
distance above/below $\bar{y}$
calculation

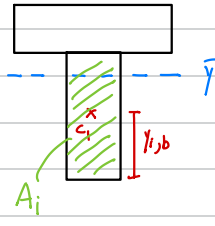
Flexural stress profile



$\bar{y}$, I for non rectangular shapes

$$\bar{y} = \frac{1}{A} \sum_{i=1}^n y_{i,b} A_i$$

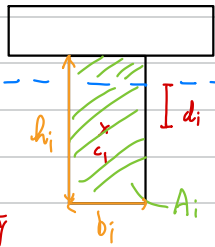
where $y_{i,b}$ is the distance of particular Area's centroid to base of shape
 A_i is the particular area



I: Apply Parallel axis theorem,

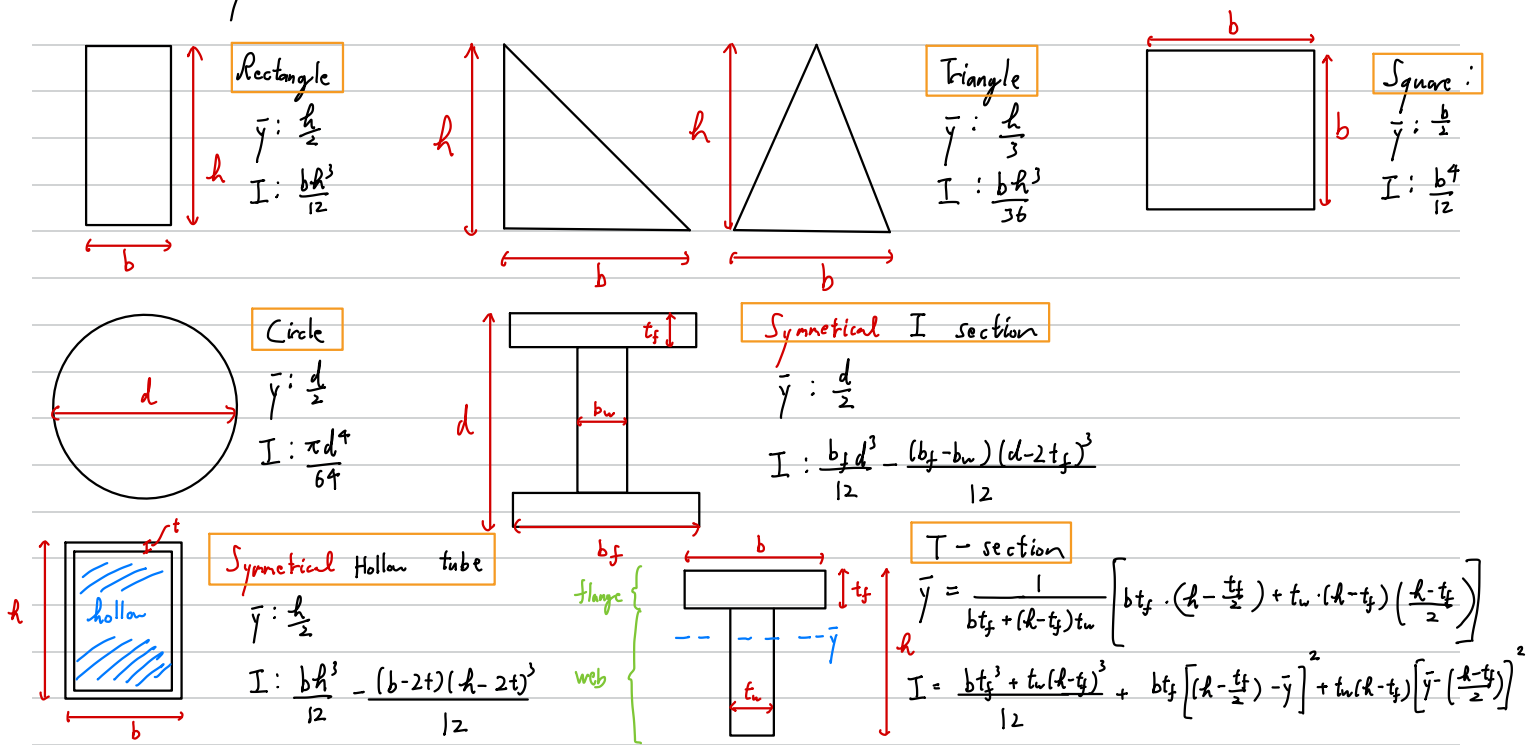
$$= \sum_{i=1}^n I_{o,i} + A_i d_i^2$$

where $I_{o,i}$ is the particular area's moment of inertia
 A_i is the particular area
 d_i is the distance from particular area's centroid to $\bar{y}$



$$I_{o,i} = \frac{b_i h_i^3}{12}$$

Common $\bar{y}$ and I_s



Moment Area Theorem 1,2

$$\phi_{\text{diagram}} = \frac{M_{\text{diagram}}}{EI}$$

MAT 1

$$\Delta \theta_{AB} = \theta_B - \theta_A = \int_A^B \phi(x) dx$$

↳ Change in slope b/w 2 points of a beam = area under the ϕ diagram

MAT 2

$$\delta_{DT} = \bar{x}_{DT} \int_D^T \phi(x) dx$$

↳ The deviation of D from tangent drawn at T is the area under ϕ diagram b/w point D and T multiplied by the distance of the centroid of the diagram to point D

Note: Centroid of ONLY that area.

↳ D → Measurement of deviation from

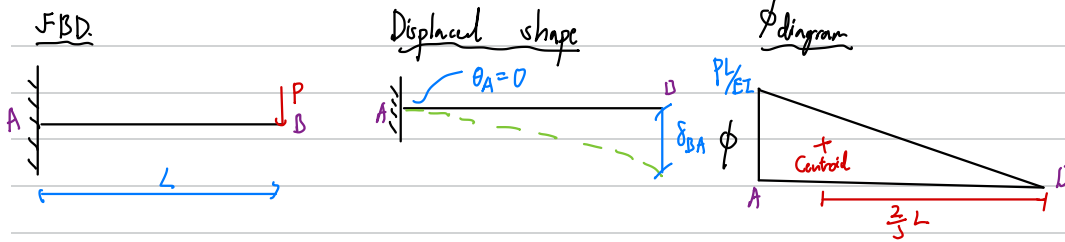
↳ T → Location of tangent



Notes: Max displacement is ALWAYS at horizontal tangent
 NOT ALWAYS at applied load

MAT Examples

Example 1, Horizontal tangent and cantilever



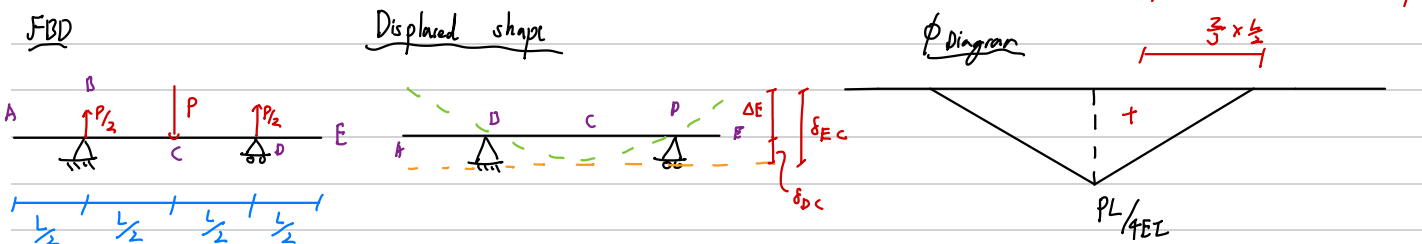
MAT 1

$$\theta_B = \frac{1}{2} \times L \times \frac{PL}{EI} = \frac{PL^2}{2EI}$$

MAT 2

$$\delta_{BA} = \left(\frac{1}{2} \times L \times \frac{PL}{EI} \right) \times \left(\frac{2}{3} \times L \right) = \frac{PL^3}{3EI}$$

Example 2, Horizontal tangent and symmetrical load



MAT 1

$$\theta_D = \frac{1}{2} \times \frac{L}{2} \times \frac{PL}{4EI} = \frac{PL^2}{16EI}$$

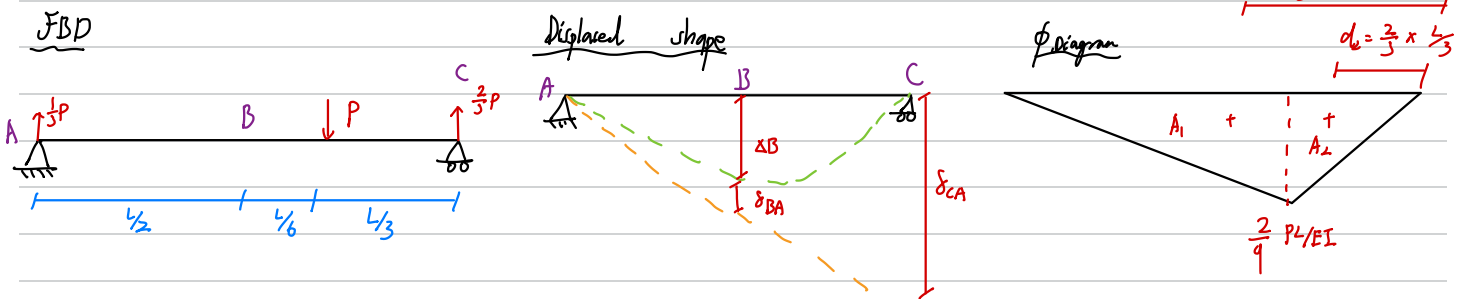
MAT 2

$$\delta_{EC} = \left(\frac{1}{2} \times \frac{L}{2} \times \frac{PL}{4EI} \right) \times \left(\frac{2}{3} \times \frac{L}{2} + \frac{L}{2} \right) = \frac{5PL^3}{96EI}$$

$$\delta_{DC} = \left(\frac{1}{2} \times \frac{L}{2} \times \frac{PL}{4EI} \right) \times \left(\frac{2}{3} \times \frac{L}{2} \right) = \frac{PL^3}{48EI}$$

$$\Delta E = \delta_{EC} - \delta_{DC} = \frac{PL^3}{32EI}$$

Example 3, \$1/6\$ horizontal tangents



MAT 1 and 2

a. $\theta_A = \frac{\delta_{CA}}{L}$

b. $\delta_{CA} = A_1 d_1 + A_2 d_2 = \left(\frac{1}{2} \times \frac{2}{3}L \times \frac{2PL}{9EI} \right) \times \left(\frac{L}{3} + \frac{1}{3} \times \frac{2}{3}L \right) + \left(\frac{1}{2} \times \frac{L}{3} \times \frac{2PL}{9EI} \right) \times \left(\frac{2}{3} \times \frac{L}{3} \right) = \frac{4PL^3}{81EI}$

c. $\theta_A = \frac{4PL^2}{81EI}$

d. By similar \$\Delta\$, $\theta_A = \frac{\delta_{CA}}{L} = \frac{\Delta_B + \delta_{BA}}{0.5L}$

e. $\delta_{BA} = \left(\frac{1}{2} \times \frac{L}{2} \times \left(\frac{2}{9} \right) \times \frac{2PL}{9EI} \right) \times \left(\frac{1}{3} \times \frac{L}{2} \right) = \frac{PL^3}{72EI}$

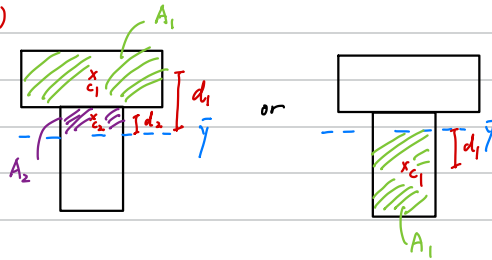
f. Rearrange and sub for d

$$\Delta_B = \frac{1}{2} \times \delta_{CA} - \delta_{BA} = \frac{7PL^3}{648EI}$$

First moment of Area, Q (and b for τ)

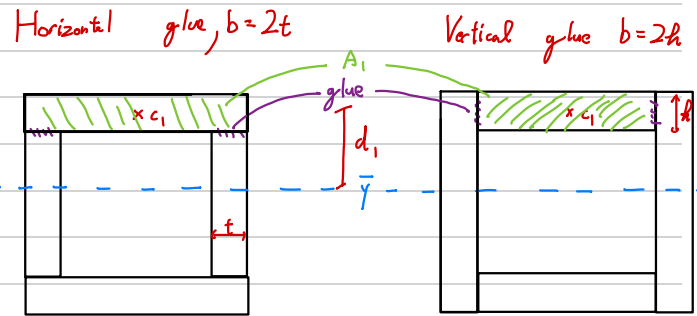
$$Q = \sum_{i=1}^n A_i d_i, \text{ from the top or bottom (same results)}$$

where, d_i is the distance from particular area's centroid to $\bar{y}$
 A_i is the particular area



$$Q_{\text{glue}} = \sum_{i=1}^n A_i d_i, \text{ top or bottom of glue}$$

where, d_i is the distance from area above/below glue's centroid to $\bar{y}$
 A_i is the area above/below glue



Shear stress

$$\gamma = \frac{V}{A}, \text{ V is shear force } \parallel \text{ to surface}$$

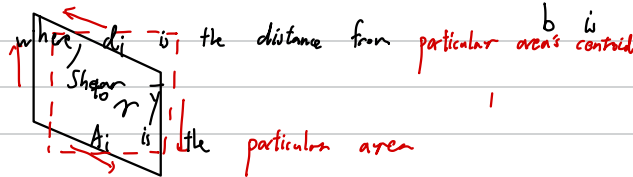
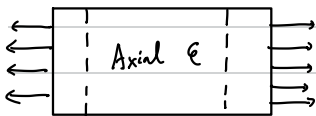
A is cross sectional area

Jourawski's equation

$$\tau = \frac{VQ}{Ib}, \text{ V is the shear force (from SFD)}$$

Q is the first moment of Area
 I is the moment of inertia
 b is the base thickness of cross-section.

Axial strain (ϵ) vs shear strain (γ)



Wood / Timber

Properties: Pros

- ↳ Light
- ↳ Stiff
- ↳ Easy to cut
- ↳ inexpensive

Cons

- ↳ rots
- ↳ Volume stability
- ↳ Higher variability
- ↳ Fire

Wood safety

- FOS: 1.5
- All properties are 5th %tile. (account for variability)
- Use $E_{0.5}$ (5th %tile) when determining failing load;
 E_{50} (Average #) when determining deflection.

↳ It is anisotropic (different properties in different orientations)

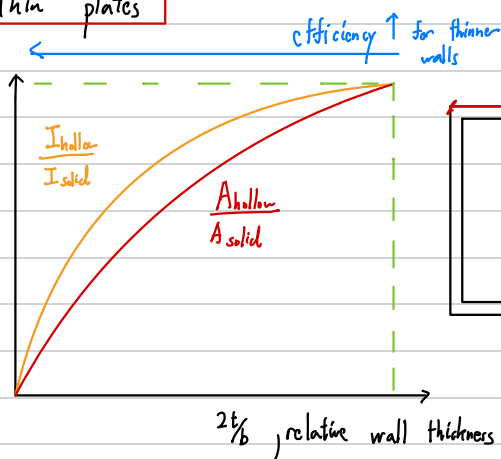
↳ Strong $\parallel$ to grain; weak $\perp$ to grain

↳ Sensitive to shear failure

Thin plates

Varying conventions

- τ_{bx} , bending stress
- τ_{mx} , longitudinal shear stress
- τ_{tx} , tensile stress $\parallel$ to grain
- τ_{pn} , compressive stress $\parallel$ to grain
- τ_{qn} , compressive stress $\perp$ to grain



$$I_{\text{hollow}} = \frac{b^4 - (b-2t)^4}{12}$$

$$I_{\text{solid}} = \frac{b^4}{12}$$

$$A_{\text{hollow}} = b^2 - (b-2t)^2$$

$$A_{\text{solid}} = b^2$$

Thin plate buckling

Plate buckling equation

$$\sigma_{\text{buckling}} = \frac{k\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2$$

k is the load specific constant
 E is Young's Modulus
 μ is Poisson's Ratio
 t is height (case dependant)
 b is base length (case dependant)

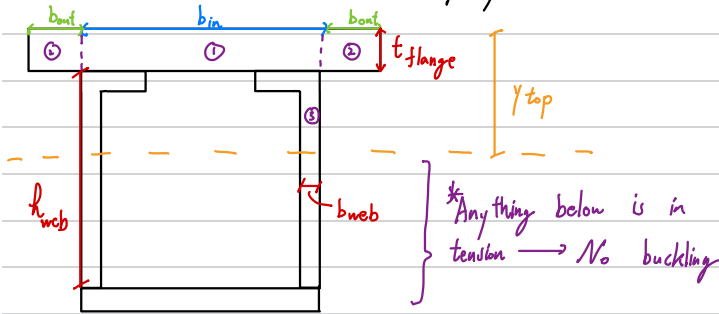
3D strain and stress

$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} \quad \sigma_x = \frac{E}{(1-\mu^2)} (\epsilon_x + \mu \epsilon_y)$$

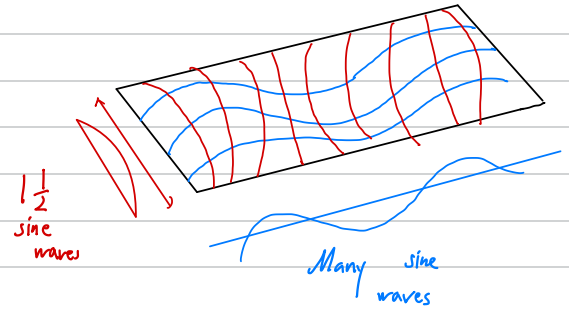
$$\epsilon_y = \frac{\sigma_y}{E} - \mu \frac{\sigma_x}{E} \quad \sigma_y = \frac{E}{(1-\mu^2)} (\epsilon_y + \mu \epsilon_x)$$

Total strain = $\epsilon_{\text{stress}} + \epsilon_{\text{poisson's ratio}}$

Consider this as an example,



2D sine wave



Case 1, all sides restrained, $k=4$

$$\sigma_{\text{buckling}_1} = \frac{4\pi^2 E}{12(1-\mu^2)} \left(\frac{t_{\text{flange}}}{b_{\text{in}}}\right)^2$$

Case 2, 1 free edge, $k=0.425$

$$\sigma_{\text{buckling}_2} = \frac{0.425\pi^2 E}{12(1-\mu^2)} \left(\frac{t_{\text{flange}}}{b_{\text{out}}}\right)^2$$

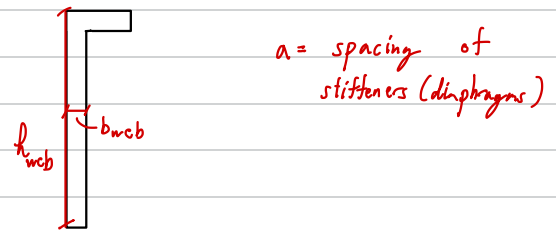
Case 3, 2 restrained edges, varying compressive stress, $k=6$

* If $y_{\text{top}} \gg t_{\text{flange}}$, $y_{\text{top}} \approx y_{\text{top}} - t_{\text{flange}}$

$$\sigma_{\text{buckling}_3} = \frac{6\pi^2 E}{12(1-\mu^2)} \left(\frac{y_{\text{top}} - t_{\text{flange}}}{b}\right)^2$$

* Consider only one web as either web may buckle independently.

Case 4, Shear buckling, $k=5$



$$\sigma_{\text{shear buckling}} = \frac{5\pi^2 E}{12(1-\mu^2)} \left[\left(\frac{b_{\text{web}}}{h_{\text{web}}}\right)^2 + \left(\frac{t}{a}\right)^2 \right]$$

Determining Failure mode of beams

Failure Mode

Tensile failure

Compressive failure

Shear failure

Shear failure (glue)

Thin Plate buckling 1

Thin Plate buckling 2

Thin Plate buckling 3

Shear buckling

Failure condition

$$\sigma = \sigma_{\text{ult}}$$

$$\sigma = \sigma_{\text{ult}}$$

$$\tau = \tau_{\text{ult}}$$

$$\tau = \tau_{\text{glue}}$$

$$\sigma = \text{Thin Plate Buckling equation 1}$$

$$\sigma = \text{Thin Plate Buckling equation 2}$$

$$\sigma = \text{Thin Plate Buckling equation 3}$$

$$\tau = \text{Thin Plate Buckling equation 4}$$

Equation

$$\sigma = \frac{M_y}{I}$$

$$\tau = \frac{VQ}{Ib}$$

$$\sigma = \frac{M_y}{I}$$

$$\tau = \frac{VQ}{Ib}$$

Find P_f failure load from BMD, SFD

Stone and Masonry / bricks

Properties:

Pro
 ↳ fire proof
 ↳ Durable
 ↳ Carving

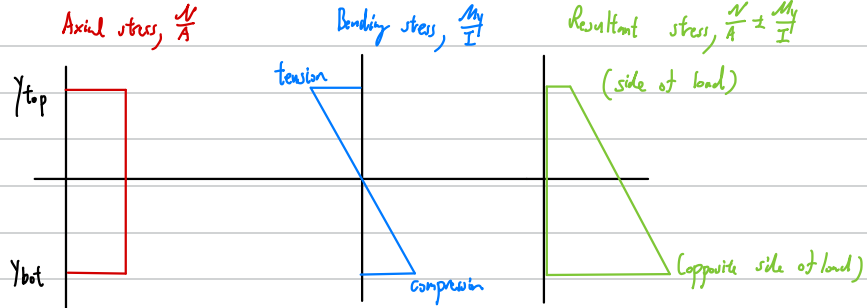
Cons
 ↳ Low tensile strength
 (Assumed 0 in calculations)
 ↳ Expensive

Stone towers

Method of combined stress

$$\sigma = \frac{N}{A} - \frac{My_{\text{left}}}{I} \quad (\text{side where load is applied})$$

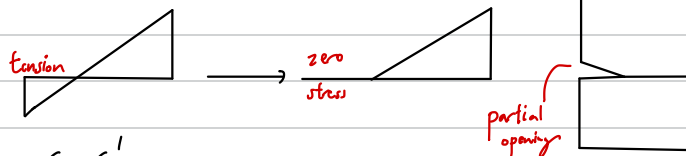
$$\sigma = \frac{N}{A} + \frac{My_{\text{right}}}{I} \quad (\text{Opposite of load})$$



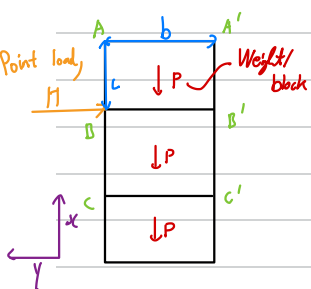
Notes:

N : Axial Force
 V : Shear Force
 M : Bending moment

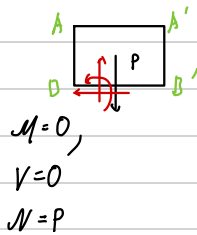
Stones cannot have "tension"



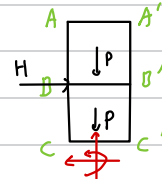
Point load



B-B':



C-C':



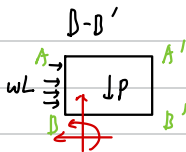
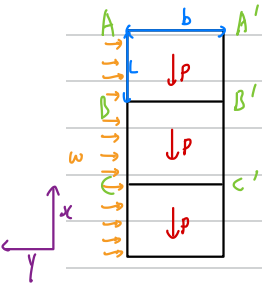
$$\sum F_x = 0, V = H$$

$$\sum F_y = 0, N = 2P$$

$$\sum M = 0, M = H(L)$$

$$qL = L^2$$

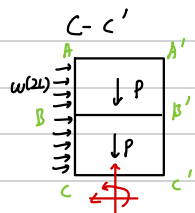
UDL



$$\sum F_x = 0, V = wL$$

$$\sum F_y = 0, N = P$$

$$\sum M = 0, M = \frac{wL^2}{2}$$



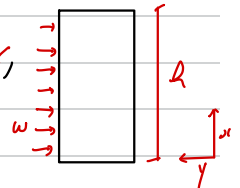
$$\sum F_x = 0, V = w(2L)$$

$$\sum F_y = 0, N = 2P$$

$$\sum M = 0, M = w(2L)(L)$$

At any point, if

- blocks have uniform density γ ,
- tower height h ,
- under uniform load w ,



Then,

$$M(x) = w(h-x)^2/2 \quad V_{\text{base}} = w \cdot h$$

$$N(x) = A \gamma (h-x) \quad N_{\text{base}} = A h \gamma$$

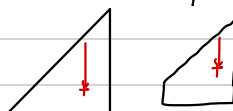
$$V(x) = w(h-x) \quad M_{\text{base}} = w h^2/2$$

Thrust line and arches

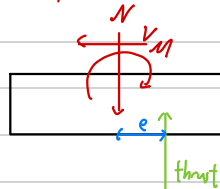


thrust line

- Resultant force is always applied at centroid of stress profile



R_x Always at centroid



$$M = N \cdot e, e \text{ is the eccentricity}$$

- Thrust line is safe if it is middle $\frac{1}{3}$ of section
- Structure collapses if thrust line is outside structure

Arches:



Concrete

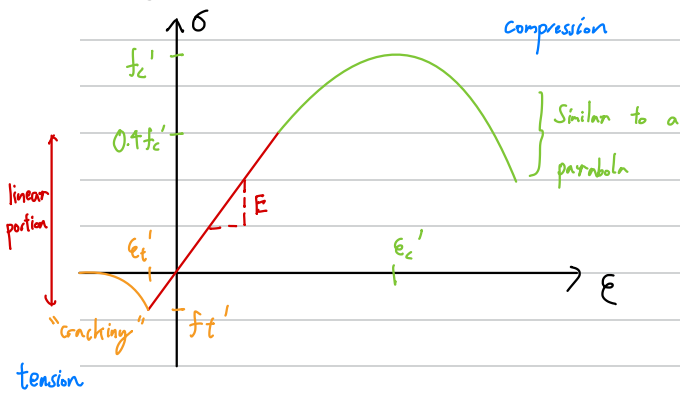
Properties:

- Pros**
- ↳ Most used construction material
 - ↳ Cheap, durable
 - ↳ local production
 - ↳ Molded to any shape
- Cons**
- ↳ Production is carbon intensive
 - ↳ Bad for the environment

Made of:

- order added
1. Aggregate (gravel) } increase volume stability
 2. Sand
 3. Cement → - Comes from grinding up lime stone and heating it to upwards of 1000°C
 4. Water
 5. Air → Deals with freezing water
 6. Admixtures → For better flowing

$\sigma - \epsilon$ Curve



Where,

f_c' is the compressive strength, f_t' is the tensile strength
Strength of cylindrical concrete sized 300x150mm tested @ 28 days of age
 ϵ_c' is the strain @ tensile crack
 ~ 20 to 30 MPa, 40 is considered normal, 100 is considered high
 $\epsilon_t' = \frac{f_t'}{E}$

Equations for estimation

$$f_t' = 0.33 \sqrt{f_c'} \quad \text{in MPa}$$

$$E_c = 4730 \sqrt{f_c'}$$

Reinforced Concrete → Concrete + rebar

Pros:

- ↳ Provides tensile capacity after concrete cracks
- ↳ Controls cracking widths, tensile deformations are distributed over multiple narrow cracks, instead of fewer wide cracks.

Rebars (Reinforcing steel bars)

- Deformed steel bars
- ↳ Helps in bonding, maintaining steel, concrete connection.



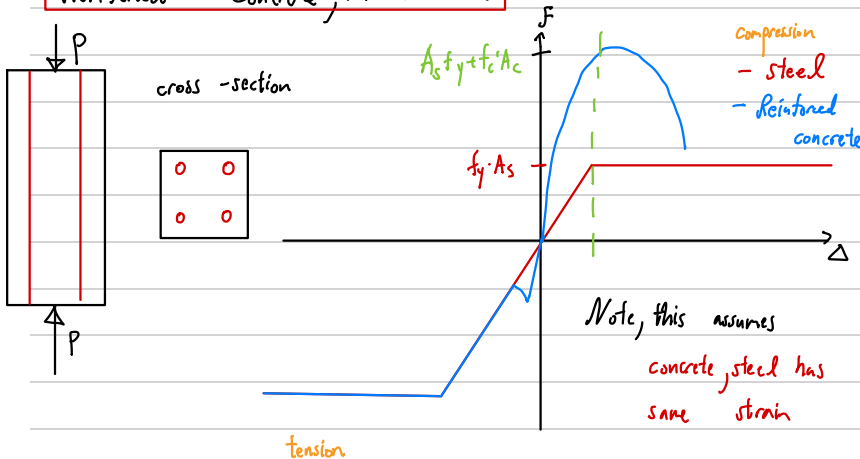
- ↳ ribs / lugs / deformations
- Initially slightly Rusty bars allows for better bonds
- Take $\sigma_y (f_y) = 400$ MPa, $E = 200,000$ MPa

Size of rebars

Name	Area (mm ²)
10M	100
15M	200
...	...
30M	700
35M	1000

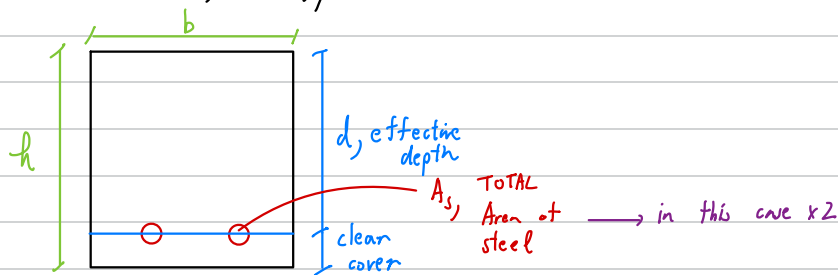
Approx. for bar diameter.

Reinforced Concrete, Axial Load



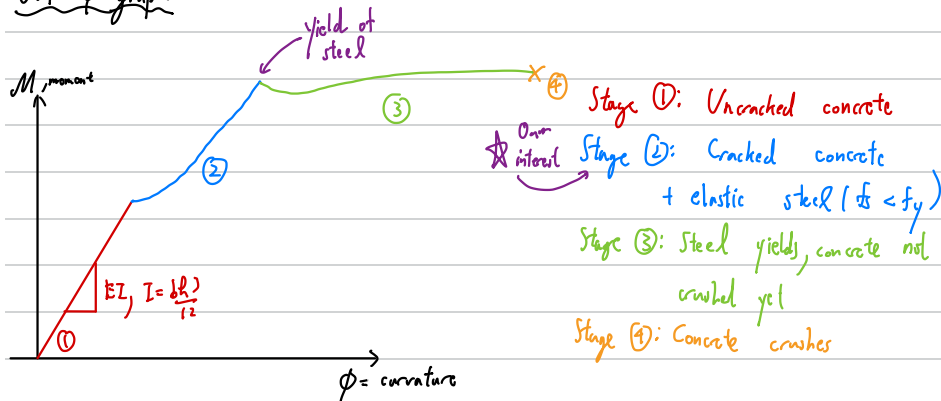
Thus max compressive capacity
 $\approx f_c' A_c + f_y A_s$
 compressive strength, concrete area of concrete compressive strength (yield) steel area of steel

Reinforced steel, terminology



Reinforced concrete, flexure

M-φ graph



Theories

-Read it yourself now.

Equations and steps required for safety

1. Find M by drawing SFD and BMD.

2. Find k by
$$\left(\sqrt{(np)^2 + 2np} \right) - np$$

where n is the modular ratio = $\frac{E_s}{E_c}$ ← Take 200,000 MPa
 E_c ← 4730 $\sqrt{f'_c}$ MPa

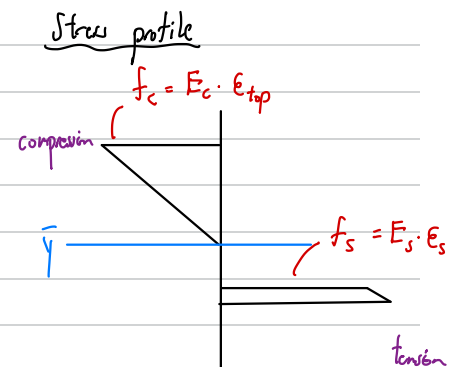
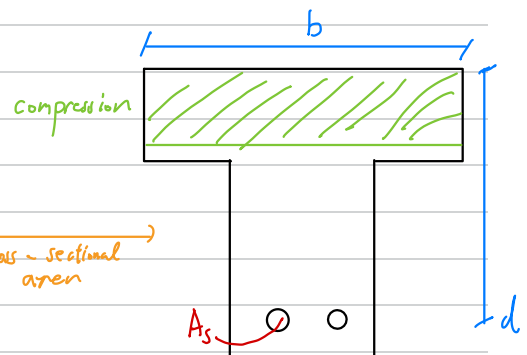
p is the % of longitudinal reinforcement = $\frac{A_s}{bd}$ ← from cross-sectional area

3. Find $jd = d \left(1 - \frac{1}{3}k \right)$ ← From step 2
 ← from cross-sectional area

4. Find applied steel stress, $f_s = \frac{M}{A_s \cdot (jd)}$ ← from step 1
 ← from cross-sectional area

5. Find applied concrete stress, $f_c = \frac{k}{1-k} \cdot \frac{M}{n \cdot A_s \cdot (jd)}$ ← from step 2
 ← from step 1
 ← from step 3
 ← from cross-sectional area

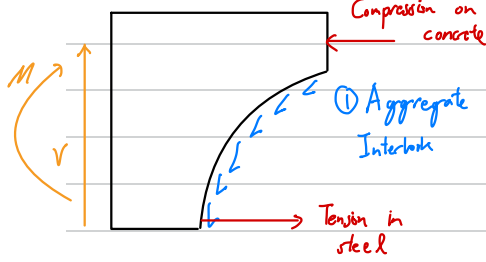
6. Allowable $f_c = \frac{f'_c}{2.0}$ ← (Typically given) and Allowable $f_s = \frac{f_y}{2.0}$ ← Typically 400



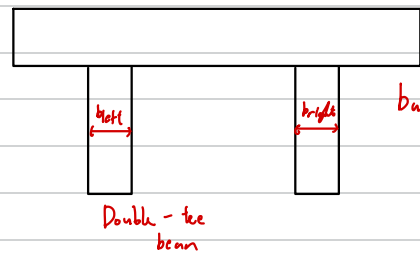
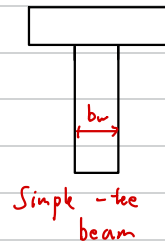
Reinforced concrete, shear

How is shear carried?

w/o reinforcement

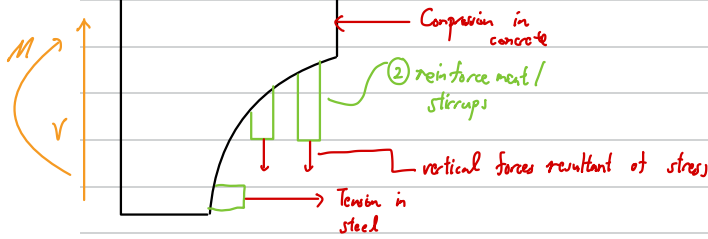


b_w , web width



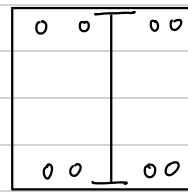
$$b_w = b_{eff} + b_{rib}$$

w/ reinforcement



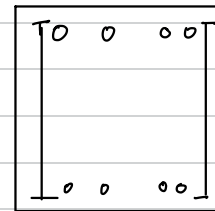
In this case, $V_{strength} = V_{aggregate} + V_{stirrup}$
 $= V_c + V_s$

A_v , area of shear reinforcement and s , spacing



$$A_v = A_{bar}$$

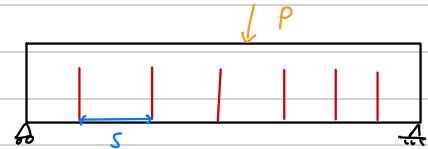
check appendix



$$A_v = 2 A_{bar}$$

Equations and steps required for safety:

1. Find $V_{applied}$ from SFD
2. Find k, ρ, n, j (same as before)
3. Find V_c , (V_s if there are stirrups)



If no (or too little) stirrups:

If stirrups are present:

$$V_c = \frac{230 \sqrt{f'_c}}{1000 + 0.9d} b_w j d$$

$f'_c \rightarrow \text{MPa}$
 $d \rightarrow \text{mm}$
 $V_c \rightarrow \text{kN}$

$$V_c = 0.18 \sqrt{f'_c} (b_w j d)$$

same units

$$V_s = \frac{A_v \cdot f_y}{s} (j d) \cot 35^\circ$$

where take 400 MPa

where

$$\frac{A_v f_y}{b_w s} < 0.06 \sqrt{f'_c} \quad (\text{or } 0 \text{ stirrups})$$

$$\frac{A_v f_y}{b_w s} \geq 0.06 \sqrt{f'_c} \quad (\text{or } 0 \text{ stirrups})$$

(Size effect $\rightarrow$ shear stress $\downarrow$ when $d \uparrow$)

4. $V_{total} = V_c + V_s$

5. Safe if $V_{total} \leq V_{max}$
 $\leq 0.25 (f'_c \cdot b_w \cdot j d)$

Not safe vice versa failure @ V_{total}

Equation and steps for design: , Note Fos is 2

1. Find $V_{applied}$ from SFD

2. Find k, ρ, n, j (same as before)

3. Check if $V_{applied} < \frac{V_{max}}{2}$

$$< 0.125 f_c' \cdot b_w \cdot jd$$

Yes $\rightarrow$ Section is too small, $\rightarrow$ resize required (??? Most likely not happening)

4. Check if $V_{applied} \leq \frac{V_c}{2}$

$$V_{applied} \leq \frac{1}{2} \left(\frac{230 \sqrt{f_c'}}{1000 + 0.9 d} b_w \cdot (jd) \right)$$

Yes $\rightarrow$ Done! Design is safe (Skip to 7)

5. Check if minimum shear reinforcement helps, i.e.

$$s = \min \begin{cases} \frac{A_v f_y}{0.06 \sqrt{f_c'} \cdot b_w} & \text{--- ①} \\ 0.63 d & \text{--- ②} \\ 600 \text{ mm} & \text{--- ③} \end{cases}$$

Then, if $V_{applied} \leq \frac{1}{2} (V_c + V_s)$

$$\leq \frac{1}{2} \left(0.18 \sqrt{f_c'} b_w jd + \frac{A_v f_y jd}{s} \cot 35^\circ \right) \xrightarrow{\text{Yes}} \text{Done! Design is safe (Skip to 7)}$$

$\underbrace{\frac{A_v f_y jd}{s} \cot 35^\circ}_{\text{from above}}$

No

6. Find the least s above minimum shear reinforcing

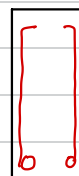
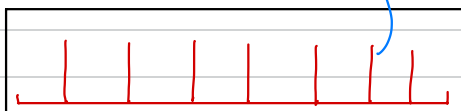
$$V_{applied} = \frac{1}{2} (V_c + V_s)$$

$$s = \frac{A_v (f_y) (jd) \cot 35^\circ}{2 V_{applied} - 0.18 \sqrt{f_c'} \cdot b_w \cdot (jd)}$$

$$2 V_a - 0.18 \sqrt{f_c'} b_w jd = \frac{A_v f_y jd \cot 35^\circ}{s}$$

7. Draw a sketch!

Example:



$A_s = ?$

$d = ?$